What Works in Teacher Policy? Evidence on Residencies, High-Stakes Evaluation, and Targeted Bonuses

Introduction

America's teachers are one of the largest and most important occupations in the United States. Roughly 5.4 million workers are employed in K-12 schools, making up 8 percent of all college-educated workers (Kraft & Lyon, 2024). The teacher profession contributes greatly to US society through educational development and the impact that teaching has on future student success (Chetty, Friedman, & Rockoff, 2014). However, organizations that provide employment, preparation, and retention for educators are struggling to meet the increasing demand for teachers. Shortages remain in many subjects and in high-poverty schools, and turnover is high, resulting in broken lines of instruction and huge costs for districts (Carver-Thomas & Darling-Hammond, 2017). Moreover, teacher quality is distributed quite unevenly across schools, with the most disadvantaged students being assigned to less experienced teachers.

For the past two decades, policymakers and administrators at the state and local level have tried multiple different strategies for addressing these issues. One of their strategies, for example, has been to create alternative certification routes so that a diverse group of individuals could enter the teaching field. Another is performance-based rating systems designed to provide constructive feedback and highlight high quality instruction. Meanwhile, compensation systems that enable recognition for good service have been developed to foster the retention of effective teachers in hard-to-fill positions. But the outcomes of implementing reforms have varied widely across the U.S. Some reforms fell short after they were implemented on a large scale (e.g., several state-based teacher rating systems created as part of Race to the Top have not achieved the level of success expected) while other efforts have succeeded in specific contexts and can provide valuable lessons to policymakers elsewhere (Bleiberg et al., 2021).

Three distinct policies will be reviewed throughout this paper, each focused on a stage of the teacher career cycle. Teacher residency programs reshape the way a teacher prepares to teach, by allowing students to gain practical experience in schools that serve as partners for a year of preparing before entering the profession. The IMPACT model, Washington D.C.'s system for evaluating teachers, provides evaluations with real-world results. Tennessee's Priority School Retention Bonus provides financial incentives to keep highly effective teachers in the state's lowest-performing schools. By analyzing the design, evidence, and replicability of each policy,

this paper draws lessons that can guide reforms aimed at strengthening teacher recruitment, development, and retention.

Policy 1: Teacher Residency Programs: Transforming Pathways into Teaching

Policy Design and Evidence of Success

Across the United States, most teachers enter the profession through university-based preparation programs. These programs typically require several years of education coursework followed by a single semester of student teaching, after which graduates assume full responsibility for their own classrooms. The structure front-loads theory and back-loads practice. By the time new teachers stand alone before students, most have accumulated no more than a single semester of supervised clinical experience (Guha et al., 2016). Research suggests this limited exposure leaves many feeling unprepared for the realities of daily instruction: a national survey found that only 40% of new teachers felt very well prepared to handle classroom management, and even fewer felt ready to differentiate instruction for diverse learners (Darling-Hammond et al., 2017). The upshot of this difference in preparation is reflected in the attrition statistics. Across their first few years, teachers who feel less prepared to teach are much more inclined to quit (Carver-Thomas & Darling-Hammond, 2017). The neediest urban schools are especially hard hit by that attrition, since they tend to recruit the least prepared teachers, who later drift away first to other jobs or leave teaching altogether.

In response to a high attrition rate, the teacher residency model was implemented before a teacher steps independently into the classroom. The framework takes a hint from medical education, where doctors train for years under supervision before eventually receiving the instruction to take care of patients. In a teacher residency, candidates spend the length of one academic year immersed in a school, co-teaching simultaneously with a seasoned mentor as they pursue their respective graduate studies. This extended immersion allows residents to experience the entire arc of a school year, from establishing routines in September to bringing instruction to closure in June. Most programs provide financial support through tuition subsidies and living stipends, and in exchange, residents commit to teaching in the partner district for multiple years following completion (Guha et al., 2016). The Boston Teacher Residency (BTR), for example, became one of the earliest and most influential implementations of this model when it launched in 2003. BTR residents spend four days weekly in Boston Public Schools classrooms working

with trained mentor teachers, and one day attending coursework at the University of Massachusetts Boston to earn a master's degree alongside their teaching certification (Papay et al., 2012). The program has since generated much of the rigorous evidence base that researchers use to evaluate whether residencies deliver on their promise of better-prepared, longer-staying teachers.

The evidence supporting teacher residencies centers on three key outcomes. First, retention is the most important of these areas, as retention remains one of the most long-standing issues within public school systems in the United States. Research shows that 8% of teachers leave the profession each year, in addition to another 8% transferring to a new school (Carver-Thomas & Darling-Hammond, 2017). This problem is compounded within low-income urban school districts, where here is an average increase of nearly 50% in teacher turnover compared to wealthier communities (Carver-Thomas & Darling-Hammond, 2017). This instability causes real harm to students who experience a drop in student achievement, loss of school culture, and lack of institutional knowledge; it can take years to restore this knowledge (Ronfeldt et al., 2013). Conversely, school districts that enter into cycles of high teacher turnover are forced to continuously absorb the costs associated with hiring and training replacement teachers, many of whom leave within a few years. It appears residency programs may lead to ending this cycle. For example, Papay et al. (2012) conducted a longitudinal study comparing the retention of BTR graduate teachers against a comparison group of teachers in the Boston Public Schools' databases. The retention difference was starkly different: Three-quarters of the BTR graduates were retained after five years, while only about one-half of the traditionally prepared teacher cohort who had been hired during that same time period remained in the district. A federal report examining residency programs from multiple states also found similar results, as residency graduates were retained in their school districts at much higher rates than their non-residency counterparts, even after controlling for the effects of differences in school placements (Silva et al., 2015). What explains these retention gains? Part of the answer may lie in how well prepared residency graduates feel when they enter the classroom. Survey evidence from the National Center for Teacher Residencies (2023) indicates that school leaders affiliated with residency programs view residency graduates as well prepared for classroom responsibilities, with over 90% of principals reporting that graduates are more effective than typical first-year teachers.

Teachers who feel competent and supported are more likely to persist through the inevitable challenges of early-career teaching.

The second outcome is teacher effectiveness. Retention gains would matter little if residency graduates were poor teachers. Here the evidence is complex but ultimately positive. Residency graduates tend to start at similar effectiveness levels as other novice teachers, but they improve more rapidly over time. For example, in a study conducted by Papay et al. (2012), BTR mathematics teachers, despite beginning somewhat less effective than their peers, were outperforming (on average) veteran teachers who had taught for six or more years by their fourth or fifth year in the classroom. The intensive clinical year seems to provide residency graduates with a stronger foundation for continued professional growth. Evidence from California supports this interpretation. Upon the evaluation of Patrick et al. (2023), 91% of residency commences in California rated their preparation as very effective or effective in general. Compared to teachers from other pathways, residency graduates reported receiving substantially more clinical hours and more frequent feedback during their training. This likely allows them to adjust to take good advantage of their own experiences once they get their licenses. The combination of high retention and accelerating effectiveness has important implications for students, as a teacher who remains in a school for ten years or more and continues to improve is more beneficial to student learning than a novice teacher who stays in a position for a few years and then leaves.

The third outcome is workforce diversity. The teaching workforce in the United States remains predominantly white. Eighty percent of public school teachers nationally identify as white while more than half of public school students do not (Ingersoll et al., 2018). This imbalance matters for a variety of reasons and has been the focus of research for decades: teacher diversity is regularly associated with positive effects on student achievement. Gershenson et al. (2022) followed North Carolina students longitudinally and documented that Black students who had as many identified teachers as they had in their elementary school years were significantly more likely to complete high school and attend college. Likewise, when taught by teachers who shared their racial/ethnic status, students scored more in standardized tests, according to Egalite et al. (2015). Diversifying the teaching workforce is, therefore, both an equity issue and an instructional approach. Residencies demonstrate measurable improvements in this regard. Nationally, 45% of teacher residents self-identify as people of color, which is twice represented from traditional university-based routes, where only 19% of people identify as this (Guha et al.,

2016). Such national patterns are reflected in states that have made a significant public investment in residency programs and intensified them in some cases. For instance, in California, 60% of educators that participated in the state's Teacher Residency Grant Program self-identified as people of color (Patrick et al., 2023). Residencies cover the program costs, which is why such program attracts more candidates identified as diverse. Tuition subsidies and living stipends allow candidates to complete a full year of intensive preparation without incurring large amounts of student debt, and without the necessity of holding down multiple jobs on the side. For candidates from financially disadvantaged backgrounds (which are disproportionately students of color), these financial supports will make a big difference in whether or not these candidates pursue a career in teaching or be precluded from pursuing a career in teaching due to the financial constraints of entering the profession.

Replicability Considerations

The rapid increase in teacher residencies in the last decade indicates that they can be replicated. More than 440 residencies were operating in the U.S. as of the 2023-24 school year (Saunders et al., 2024). Many were seeded by federal Teacher Quality Partnership grants. Several states have also made sizable investments of their own. California dedicated more than $670 million over five years to teacher residencies. Tennessee, Pennsylvania, and other states have implemented their own as well (Saunders et al., 2024). The large number of residency programs being created represents the ability of the model to spread beyond its original site in Boston. However, the quality of residency programs is very different. Some residency programs have demonstrated the strong retention rates and the improved performance associated with them. Others have not been so successful. Therefore, it is important for a policymaker to know what conditions are necessary for success in order to determine if this is an approach they want to take.

The biggest impediment to providing residency opportunities is funding. Training one resident costs between $35,000 and $60,000, including stipend, subsidized tuition, mentor compensation, and administrative overhead (Worley & Zerbino, 2023). Although states or federal grants can be used to fund these expenses initially, they are typically not permanent solutions. The true test of sustainability is after a grant has expired. A number of programs in California have had difficulty transitioning between funding cycles, and therefore have either temporarily closed or were at least struggling (Fitz & Yun, 2024). The Tri-County Teacher Residency, as an

example, closed down completely for the 2023-24 school year because the program was waiting for its next round of funding to be approved. All of the programs that continued past their initial funding period have one thing in common: the districts that partner with them provide some form of local financial support. The Kern Urban Teacher Residency, for instance, gets between $750,000 and $1.2 million a year in funding from the Bakersfield City School District through the Local Control Funding Formula (Fitz & Yun, 2024). If a program does not have the monetary support of the districts that participate, they will always be at risk of closing down when outside funding ends. Thus, when policymakers develop residencies or expand already existing residencies, they must begin efforts to create buy-in from the districts and develop sustainable forms of local funding for residencies beginning the first day of the residency program and not wait until grant funding expires.

In addition to funding, the quality of mentoring is another important factor in determining the success or failure of a residency program. The clinical year is only valuable if the residents are provided meaningful direction from an experienced mentor. What does that direction look like in practice? A study conducted across several states on induction and mentoring of new teachers conducted by Goldrick and colleagues (2012) identified many of the characteristics associated with strong outcomes for new teachers. Successful residency programs include protected time for residents and mentors to routinely meet and have structured interaction; the New Teacher Center recommends between 1.25 - 2.5 hours of structured interaction per week. Successful residency programs offer training to new teachers regarding how to observe instruction, how to give constructive feedback to adults, and how to support adult learners. When these elements of a successful residency are present, new residents report feeling better prepared to teach, and the retention rates for the residents tend to be higher (Goldrick et al., 2012). However, not all residency programs within the state of California devote the same level of resources or energy to mentoring. Some residency programs have limited funding and pressure to admit more candidates to the residency, and therefore provide mentors without the adequate training, or dedication of the time that is required to mentor them effectively, and in many cases, mentoring becomes a formality instead of a developmental opportunity for the mentor and the mentee. As residency programs continue to expand, providing mentorship to their residents will continue to become a greater challenge. Districts will only have a certain number of teachers who possess the ability, experience and desire to serve as effective mentors to their residents. If

districts continue to increase the number of residents enrolled without increasing the pool of effective mentors to work with the residents, they will dilute the quality of the clinical experiences for the residents. For this reason, when policymakers develop residency programs or expand residency programs, they must take into consideration the mentoring capacity of the district, and, as a result, policymakers should begin investing in developing effective mentors at the same time they are recruiting residents.

Third, the financial support residents receive during their training is also significant. A resident cannot earn a full teacher's salary while they are working four days a week in the classroom of an experienced teacher. For candidates without savings or family resources to draw upon, this creates a serious barrier. Individuals from lower-income backgrounds face the greatest difficulties, and these individuals are disproportionately people of color. Because one of the central goals of residency programs is to diversify the teaching workforce, inadequate stipends work directly against program objectives. Some promising candidates never apply because they cannot afford to. Others start the program but drop out when financial pressures become unsustainable. To fulfill the diversity potential of the residency model, programs must offer stipends sufficient for residents to cover basic living expenses without taking on crushing debt or working so many outside hours that their clinical learning suffers.

Policy 2: DCPS IMPACT: High-Stakes Evaluation as a Lever for Improvement

Policy Design and Evidence of Success

Teacher evaluation has long been a weak link in American public education. For much of the twentieth century, most districts relied on simple systems that did little to differentiate among teachers or to support their improvement. A landmark 2009 report titled *The Widget Effect* documented the extent of this problem. After examining evaluation practices in twelve districts across four states, researchers found that in systems using binary rating categories, fewer than 1% of teachers received unsatisfactory ratings (Weisberg et al., 2009). The impacts of this initiative were severe, as teachers had similar overall scores regardless of performances, which meant the evaluation was not conducive to supporting teachers' development of professional skills. Administrators had little background information to base their decisions on regarding tenure, promotions, and dismissals; therefore, less-than-effective teachers could remain in the classroom for a long time, and on the other hand, there was little recognition for effective

teachers (Weisberg et al., 2009). Ultimately, it was the students enrolled in schools with the highest levels of need who experienced the greatest cost associated with having a greater number of teachers without competency and fewer resources to support performance (Glazerman et al., 2010).

Beginning in 2009, D.C. Public Schools began to implement the IMAPCT initiative to change the status quo. The launch of IMPACT by Chancellor Michelle Rhee has been considered one of the most ambitious teacher evaluation reforms in America. The IMPACT program measures teachers using a multiple-faceted approach. Approximately 50% of a teacher's annual evaluation is derived from classroom observations conducted by the principal and certified staff observers trained in the Master Educator Process. These observers assess classrooms on multiple visits throughout the year using assessments based on the rubric called the Teaching and Learning Framework. The rubric explains what teaching looks like in terms of dimensions such as the teachers' ability to clearly explain lesson content and check for student understanding. The remaining portion of the teacher's rating is determined by student achievement and, specifically, by student value-added assessments that estimate the learning impact of teachers beyond student characteristics and prior student achievement. IMPACT was designed to improve teacher performance not just by evaluating teachers' performance, but by having real consequences tied to how teachers were rated. Teachers rated Highly Effective may receive sizeable incentives sometimes as much as $25,000. Being rated Highly Effective also gives them faster salary increases and higher odds for promotion into more senior leadership positions. Conversely, those rated Ineffective face real consequences. All teachers rated as Ineffective must have a requirement for developing a formal improvement plan. If a teacher is rated as Ineffective for two consecutive years, that teacher will be subject to termination of their employment with the school. Regardless of this, the district has consistently and aggressively implemented these consequences, and this aggressive follow-through is a significant factor determining the IMPACT program's success (Dee & Wyckoff, 2015).

The rationale for using the IMPACT program to measure teacher quality is based on two outcomes from the implementation of the IMPACT program. One is heightened teacher quality by what has been highlighted in studies as a selection effect. High-stakes evaluation changes who teaches in a district by encouraging low performers to leave and retaining or attracting stronger teachers. Dee and Wyckoff (2015) found that teachers who received low IMPACT ratings were

significantly more likely to exit the district voluntarily, with departure rates increasing by 11 percentage points among this group. More importantly, the teachers who replaced these departures were, on average, more effective than those who left (Adnot et al., 2016). The teaching workforce, in other words, shifted toward higher quality. This pattern distinguishes IMPACT from most other evaluation reforms implemented during the same period. When Bleiberg et al. (2021) analyzed teacher evaluation policies adopted across the country after Race to the Top, they found that most produced no detectable gains in student achievement. The reason was implementation. Most districts adopted evaluation systems that included rigorous-sounding components, such as multiple rating categories and student achievement measures, but failed to attach meaningful consequences. IMPACT, along with a handful of other exemplar systems, succeeded precisely because it enforced consequences consistently.

The second outcome is improved performance among teachers who remained in the system. High-stakes evaluation can motivate teachers to strengthen their practice, particularly when both accountability pressure and financial rewards are present. Dee and Wyckoff (2015) found evidence of this incentive effect at both ends of the performance distribution. Teachers who received low ratings but chose to stay improved their effectiveness by 0.27 standard deviations in subsequent years. Meanwhile, teachers who qualified for performance bonuses by achieving high ratings also continued to improve, gaining an additional 0.24 standard deviations. The educational research sector often associates the above stated effect sizes as meaningful in terms of significance (Hill et al., 2008). More critically, gains of this nature tend to compound annually. A slowly improving educator adds exponentially more knowledge to students than one who has stagnated throughout their career. Dee and Wyckoff (2015) calculated the corresponding student performance increases from IMPACT as 0.15 standard deviations, a significant improvement for a large urban district with thousands of students.

Replicability Considerations

IMPACT's success can be replicated, but some obstacles make the journey difficult.

The challenge that took the longest was maintaining political will so that true high-stakes implementation can be enforced. IMPACT was happening at an unusual situation in the Washington, D.C. setting. The Chancellor of D.C., Michelle Rhee, had support from the mayor, Adrian Fenty, and she was willing to keep clashing with the teachers' union in order to gain more advantages for her reforms. Her entire career was full of protests, media attention, contract

negotiations, and much other drama. Few district leaders have the same support from their mayors or school boards as the case was in Washington, D.C., and even fewer are willing to continue on this path of disruption. The experiences of other states demonstrate that this is not easy to do. For most states, they adopted teacher evaluation reforms after Race to the Top, which seemed rigorous in terms of theory and mechanics. However, in practice, many districts watered down these systems, or they did not meet the requirements. According to Bleiberg et al. (2021), the majority of state-level teacher evaluations did not affect students' outcomes. The lesson taught to policymakers was simple: consequences must be enforced by the leadership and repeated over several years to obtain results. Otherwise, even the best plans will underperform.

The second issue relates to the development of technical infrastructure that is essential for high-stakes evaluation. IMPACT relies on two technical capacities: The first refers to classroom observations that are reliable and consistent. IMPACT uses master educators who are trained meticulously on the observation rubric, and the district has maintained continuous calibration so that different observers rate the lessons similarly. As long as teachers believe that the observations are conducted fairly, they will generally accept the evaluation results, even if they receive unfavorable feedback (Donaldson & Papay, 2014). Conversely, if teachers feel that the evaluations are arbitrary, the evaluation system as a whole will lose credibility. The second capacity relates to accurate value-added measures. This means that there must be individual longitudinal test data from students that is linked to teachers, as well as statistical expertise to develop and interpret these results. Many districts do not have these data systems or building and staffing to maintain them. Developing such an infrastructure is a process that occurs over many years, and thus, districts that try to implement high-stakes evaluation without the depth to support it will often create systems that generate resentment, rather than improvement.

A third consideration is ongoing refinement. Because evaluation systems in high-stakes situations are implemented in complex environments, changes over time can lead to unintended outcomes. IMPACT has changed since it originally launched in 2009. Early on, there was an imbalance in the way that the evaluation system weighted value-added measures and a teacher's contribution to student growth. Some teachers expressed concern that elements outside of their control, such as student mobility and class composition, could negatively impact their evaluation ratings. As a result, the District restructured the formula to place a greater emphasis on evaluations through observation (Dee & Wyckoff, 2015). District leaders also revised the

observation rubric as a result of feedback from teachers and principals, and provided additional training for observers to ensure that they used the rubrics uniformly when applying evaluation standards. The multiple iterations of IMPACT have continued to provide improved teacher quality and student achievement. The takeaway from this example is a reminder that evaluation is a process that must be based on feedback and continue to evolve as more details are discovered. The responsibility of a policymaker is to create alternate means to hear the voices of teachers as well as administrators regularly. As the creation and finalization of these means occur, increasingly more feedback and insights are expected to be received through the utilization of the methods, thus likely revealing unexpected situations. Therefore, to maintain the integrity of the IMPACT system and to improve the performance of the educators, it requires addressing the issues from the revealed situations through variable assessment calculations and assessment rubrics or modifications of observer training.

Policy 3: Tennessee Priority School Retention Bonus: Targeted Compensation for Equity

Policy Design and Evidence of Success

The preceding two policies establish methods to effectively prepare and meaningfully evaluate teachers. The third policy changes the focus to addressing a different issue pertaining to distributing qualified teachers equally among schools that educate different groups of students. A wealth of evidence exists demonstrating that students in high-poverty schools do not have the same access to effective teachers as do students in higher-income areas. Lankford et al. (2002) studied the distribution of teachers in New York State and concluded that schools with primarily low-income students and students of colour employ less qualified, less experienced, and lower certified teachers. Additionally, Goldhaber et al. (2015) confirm that the patterns identified by Lankford et al. exist across multiple states and are indicative of the structural nature of the local teacher labour market. The impact on students is significant, since the most important factor influencing student achievement in a school is teacher effectiveness (Rivkin et al., 2005); inequitable distribution of teachers creates additional inequities for students who already face barriers to academic achievement.

To understand why this inequity is present, we need to look at how teacher labor markets work. Teachers, as with any other professional, respond to conditions in their workplace when making decisions about their careers. Schools serving high-poverty populations tend to be

difficult environments to work in, where there is higher student mobility, fewer available resources, and more concentration of academic and behavioral challenges. When presented with otherwise equal job opportunities, many teachers gravitate to placements where the work is more manageable. Conventional compensation structures do nothing to counter this preference. Teacher compensation nearly everywhere in the United States follows path-dependent wage schedules based on experienced years in a role and educational credentials. A teacher earns a base salary irrespective of working in a school with solid help networks or where the school grapples with ongoing instability. Hanushek et al. (2004) showed that teachers were more likely to exit from their Texas schools as they became more disadvantaged, even after controlling for salary differences. It is evident that, without focused financial incentives, hard-to-staff schools should expect to keep losing effective teachers from being pressured to leave underserved schools for environments with much less demanding work.

To support its challenge with the increased complexity of retaining quality educators at Priority Schools, Tennessee's Priority School Retention Bonus Initiative offers a $5,000 annual retention bonus to selected teachers employed at Priority Schools designated as those within the bottom 5 percent of the state relative to their academic achievement while also being rated as Highly Effective within the state of Tennessee's Teacher Evaluation System. Being paid an $5,000 bonus, representing 8 to 10% of the average teacher salary, provides fairly substantive financial recognition for choosing to stay in a difficult supporting environment. Tennessee based this intervention on its existing evaluation infrastructure, first developed under Tennessee's Race to the Top participation. By using a well-defined system for defining both Priority Schools and Highly Effective teachers, the policy avoided the administrative burden that hampered competing reform initiatives.

Two specific outcomes provide the rationale that supports the implementation of this policy. The first is the higher retention rate of effective teachers in Priority Schools. Springer et al. (2016) documented significant increases in teacher retention rates of effective teachers at Priority Schools that were receiving retention bonuses. This finding is particularly applicable in Dallas Independent School District, where the Accelerating Campus Excellence program provides salary supplements of between $6000 and $10000 for teachers at hard-to-employ schools. Morgan et al. (2023) found that these salary supplements had significant effects on some schools in terms of being able to decrease the percentage of teacher turnover, as well as increase

the academic improvement of their students. The retention effects provide an important model for the policies that tackle issues of educational justice. While schools serving high-poverty student populations have historically been unable to compete for effective teachers, the continuing cycle of teacher turnover has created and contributed to widening achievement disparities between advantaged and disadvantaged students (Goldhaber et al., 2015). Through providing Priority Schools the capacity to retain their most effective teachers, Tennessee's bonuses begin to address this long-standing issue of educational injustice. Students in the state's lowest-performing schools gained access to the kind of instructional continuity and expertise that students in more affluent schools have long taken for granted.

The second outcome is favorable cost-effectiveness relative to other approaches to improving teacher quality. Swain et al. (2019) estimated the impact on student achievement when highly effective teachers were retained rather than replaced. Students in classrooms with retained effective teachers gained between 0.08 and 0.12 standard deviations in reading achievement compared to students who would have been taught by typical replacement teachers. These gains may appear modest in isolation, but they accumulate when students experience effective teaching consistently over multiple years. Moreover, Springer et al. (2016) translated these achievement differences into projected lifetime earnings using established estimates from the economics of education literature. They estimate that a single year of exposure to a teacher one standard deviation more effective can increase the total lifetime earnings of a class of 25 students by more than $500,000 in net present value. A retention bonus of $5,000 provides a significant financial incentive that has the potential to yield much higher returns than its initial investment.

Replicability Considerations

Across other states and districts, retention bonuses of varying types are also implemented, which implies that the targeted approach by Tennessee can be applied to other settings as well. To illustrate, Hawai'i increased the salaries of all special education teachers by $10,000 in 2020 and further paid an additional $8,000 for working in hard-to-staff schools. Such incentives were successful in decreasing special education vacancies, where large drops were recorded in the hardest-to-staff schools (Theobald et al., 2023). Similarly, the Accelerating Campus Excellence program of Dallas Independent School District utilized performance-based salary supplements of $6,000 to $10,000 as a way of stabilizing staffing in low-performing schools, thus positively

affecting experience teachers' retention as well as the students' outcomes (Morgan et al., 2023). These experiences further show that financial incentives have the potential to affect teachers' effectiveness and where they choose to be employed, although local conditions affect the specific design and amounts of the bonus. Finally, there are two important considerations in terms of implementing the policy, which will be further elaborated below.

The first is a credible evaluation system that is able to identify the best teachers. Tennessee's bonus relies, in part, on the state's ability to differentiate between top performers and average teachers. The state had spent several years developing and revising its evaluation system prior to its release in conjunction with the retention bonus. Without this foundation in its evaluation process, a targeting of the bonus towards Highly Effective teachers will not work well. If an evaluation system assesses nearly all teachers as effective, as was the case prior to several rounds of reforms, then the bonuses will be rendered powerless with the bonus functioning more as a salary supplement for all teachers in designated schools (Weisberg et al., 2009).

The second is a long commitment to stable parameters of the program. Teachers make their careers based on future expectations. A teacher who is trying to decide whether to remain at a Priority School will need to feel comfortable with the bonus continuing to be available and offering consistent criteria for entry. If the program haphazardly shifts funding after one or two years, or the definition of Priority Schools continues to be redefined at whim, then the bonus cannot be integrated into the long-term planning of teachers. Tennessee sustained an organized set of eligibility criteria and consistent funding for multiple years, allowing the incentive to matter in teachers' decisions. To promote similar success, policymakers should avoid short-term funding periods from the outset. They should also adjust bonus amounts to local salary levels and cost of living. Where teacher salaries are higher or cost of living is increased, $5,000 may not be enough to persuade the retention choice, and larger sums may be needed to create the same effect.

Cross-Policy Analysis: Lessons for Policymakers

Policies targeting three different stages of a teacher's career share similar characteristics. Looking at them together gives us real lessons learned for those interested in making reforms in teacher recruitment, development and retention.

The first lesson is that successful policy examples solve certain problems having well-defined boundaries in contrast to an overall broad transformation. Each of the three models addressed specific barriers along the path of becoming a teacher. The residency program model addressed the gap that exists between coursework and actual classroom experiences for teacher candidates, by providing clinical experience for one full year. IMPACT, provided the means for schools to differentiate among teachers in terms of their ability, via a robust multi-faceted system of evaluation. The Tennessee bonus system addressed the issue of equitable distribution of quality teachers to high-needs schools through financial incentives to retain teachers in those positions. None of the three policy approaches attempted to make extensive reform to the entire teacher pipeline simultaneously; rather, they recognized a specific point of leverage along the way to create success.

The second lesson is that policymakers should offer incentives that are large enough to create real change to the teacher's career. Financial rewards must be large enough to be able to create a positive shift in a teacher's career decision-making process. For instance, highly effective teachers received a bonus greater than $25,000 through IMPACT, which was more than enough to influence the decision-making of the effective educators in the school district between pursuing other career options or remaining in their school setting. Tennessee's annual $5,000 bonus is approximately 8-10% of an average teacher's salary, and is therefore sufficient to provide a reason for teachers to remain in challenging school environments. Residencies serve as an education support by providing tuition assistance and living allowance in order to sustain a year-long clinical practice for career candidates. For each situation, the policymakers set clear behavioral targets that were run according to the level of certain incentives they wanted to effect different.

The third lesson is that policies that have stability allow teachers to make long-term career decisions. Teachers plan for careers for years to come and even decades ahead. If policies are shifting unpredictably or never stay after the experimental stages, then educators cannot embrace them in their plans. Tennessee maintained a consistent administrative process for its retention bonus over multiple years, allowing teachers to make an informed choice regarding whether or not to remain in a Priority School. The consistency of the IMPACT evaluation system has endured through changes in the local political landscape, repeatedly demonstrating the system's commitment to teachers and to reforming and improving teacher quality. Furthermore,

the Boston Teacher Residency Program has been in operation for 20 years, creating a body of institutional knowledge about the development and implementation of the residency model—and they haven't stopped there, taking the data they gather and using it to improve and refine the model. When building a change in teacher quality, policymakers need to commit to putting their initiatives in place long enough for teachers to work it into their career planning.

Conclusion

Teacher residencies, high-stake evaluation, and retention bonuses have been proven effective in modeling how to develop a strong workforce of teachers. Collectively, these three policies provide policymakers with three important lessons: 1) to identify a specific workforce issue and provide solutions to that problem, 2) to ensure that adequate financial resources are allocated to deal with this problem, and 3) to be persistent over time. It is crucial that policymakers keep in mind these lessons above in order to establish a teaching workforce capable of serving the needs of all learners equitably and effectively.

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